

Appendix - Max Kropinak

1 - R_0

```
1 syms sigma mu gamma_c alpha delta omega beta_d beta_e S_star p
2
3 F = [p*(mu+gamma_c), beta_d*S_star, beta_e*S_star;
4       0,0,0;
5       0,0,0];
6
7 V = [sigma+mu+gamma_c, 0, 0;
8       -sigma,alpha+mu+gamma_c,0;
9       0,-omega,delta];
10
11 FV_inv = F*inv(V);
12
13 R_0_raw = simplify(eig(FV_inv));
14 R_0 = simplifyFraction(R_0_raw(3))
```

R_0 =

$$\frac{\delta p \gamma_c^2 + 2 \delta p \gamma_c \mu + \alpha \delta p \gamma_c + \delta p \mu^2 + \alpha \delta p \mu + S_{\text{star}} \beta_d \delta \sigma + S_{\text{star}} \beta_e \omega \sigma}{\delta (\alpha + \gamma_c + \mu) (\gamma_c + \mu + \sigma)}$$

```

1 % CWD Project - Max Kropinak
2
3 % Parameters
4 r = 0.6;
5 K = 1000;
6 mu = 0.1;
7 beta_d = 0.0005;
8 beta_e = 0.00025;
9 sigma = 0.5;
10 alpha = 0.8;
11 omega = 1;
12 delta = 0.1;
13 gamma_c = 0;
14 p = 0.5;
15
16 % Initial Conditions
17 S0 = 950;
18 E0 = 35;
19 I0 = 10;
20 P0 = 5;
21 y0 = [S0; E0; I0; P0];
22
23 % Time interval
24 tspan = [0 100];
25
26 % Solve system
27 [t,y] = ode45(@(t,y) cwd_model(t,y,r,K,mu,beta_d,beta_e,sigma,alpha,omega,delta,gamma_c,p), tspan, y0);
28
29 % Extract variables
30 S = y(:,1);
31 E = y(:,2);
32 I = y(:,3);
33 P = y(:,4);
34 N = S + E + I;
35
36 % Plot trajectories
37 figure;
38 plot(t,S,'LineWidth',2); hold on;
39 plot(t,E,'LineWidth',2);
40 plot(t,I,'LineWidth',2);
41 plot(t,N,'LineWidth',2);
42 xlabel('Time (in years)');
43 ylabel('Population');
44
45 title('CWD Model - Baseline Trajectories');
46
47 legend('S','E','I','N','Location','best');
48 grid on;
49 % -----
50 % CWD ODE System
51 function dydt = cwd_model(~,y,r,K,mu,beta_d,beta_e,sigma,alpha,omega,delta,gamma_c,p)
52 S = y(1);
53 E = y(2);
54 I = y(3);
55 P = y(4);
56 N = S + E + I;
57
58 dS = r*(S+(1-p)*E)*(1-N/K)-(beta_d*I+beta_e*P)*S-(mu+gamma_c)*S;
59 dE = (beta_d*I + beta_e*P)*S + p*r*E*(1-N/K)-(sigma+mu+gamma_c)*E;

```

```
60 dI = sigma*E-(alpha+mu+gamma_c)*I;  
61 dP = omega*I - delta*P;  
62  
63 dydt = [dS; dE; dI; dP];  
64 end
```

3

```
1 % -----
2 % Iteratively varying the culling rate
3
4 culling_test = [0, 0.05, 0.1, 0.25, 0.5]
5 tags = {'r', 'g', 'b', 'm', 'w'}
6
7 % Total population after culling
8 figure; hold on;
9 for j = 1:length(culling_test)
10     gamma_c = culling_test(j);
11     [t,y] = ode45(@(t,y) cwd_model(t,y,r,K,mu,beta_d,beta_e,sigma,alpha,omega,delta,
12         gamma_c,p), tspan, y0);
13     N = y(:,1) + y(:,2) + y(:,3) ;
14     plot(t, N, tags{j}, 'LineWidth', 2);
15 end
16
17 yline ( 0.75*K, '--r', 'LineWidth', 2);
18 xlabel('Time (in years)');
19 ylabel('Total Population N');
20 title('CWD Model - Varying Culling Rates');
21 legend('\gamma_c=0', '\gamma_c=0.05', '\gamma_c=0.1', '\gamma_c=0.25', '\gamma_c=0.5', '0.75K',
22     'Location','best');
23 grid on;
24
25 % Infected population after culling after culling
26 figure; hold on;
27 for j = 1:length(culling_test)
28     gamma_c = culling_test(j);
29     [t,y] = ode45(@(t,y) cwd_model(t,y,r,K,mu,beta_d,beta_e,sigma,alpha,omega,delta,
30         gamma_c,p), tspan, y0);
31     plot(t, y(:,3), tags{j}, 'LineWidth', 2);
32 end
33
34 xlabel('Time (in years)');
35 ylabel('Infected Population I');
36 title('CWD Model - Varying Culling Rates');
37 legend('\gamma_c=0', '\gamma_c=0.05', '\gamma_c=0.1', '\gamma_c=0.25', '\gamma_c=0.5', '
38     Location','best');
39 grid on;
```

```

1 % -----
2 % Calculating the optimal culling rate
3
4 gamma_test_values = 0:0.001:0.3;
5 infection_prevalance = zeros(1, length(gamma_test_values));
6 N_final = zeros(1, length(gamma_test_values));
7
8 for j = 1:length(gamma_test_values)
9     gamma_c = gamma_test_values(j);
10    [t,y] = ode45(@(t,y) cwd_model(t,y,r,K,mu,beta_d,beta_e,sigma,alpha,omega,delta,
        gamma_c,p), tspan, y0);
11
12    % Store the long term trajectory for all populations
13    S_end = y(end,1);
14    E_end = y(end,2);
15    I_end = y(end,3);
16    N_end = S_end + E_end + I_end;
17
18    % Put results in the N_final
19    N_final(j) = N_end;
20    % If population didn't go extinct, find the prevalence for goal 1. If
21    % not, it should be 0.
22    if N_end > 0
23        infection_prevalance(j) = I_end / N_end;
24    else
25        infection_prevalance(j) = 0;
26    end
27 end
28
29 % Plot the proportion of infected deer based off the culling rate
30 figure;
31 plot(gamma_test_values, infection_prevalance, 'LineWidth', 2);
32 xlabel("\gamma_c");
33 ylabel("Infection Prevalence : I*/N*")
34 title("Infection Prevalance vs Culling Rates")
35 grid on;
36
37 % Plot the deer population based off the culling rate
38 figure;
39 plot(gamma_test_values, N_final, 'LineWidth', 2);
40 yline(0.75*K, '--r', 'LineWidth', 2);
41 xlabel("\gamma_c");
42 ylabel("Total Population N*");
43 title("Long-Term Population vs Culling Rates");
44 legend('N*', '0.75K', 'Location', 'best');
45 grid on;

```